COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010 - Data Structures & Algorithms	**SECTION:**	CPE21S4
DATE PERFORMED:	October 9, 2026	**DATE SUBMITTED:**	October 13, 2026
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ACTIVITY NO. 11.1 Planning

1. PROCEDURE OUTPUT

General Instructions:

1. Form your group for the project and submit the names in a 1/4 sheet of yellow paper.
2. Write down your 3 proposed systems to develop.
3. Meet with your group and decide what type of SDLC you will use to create a gantt chart and discuss the task responsibilities.
4. What possible 10 questions will you ask to find out how the business process works for one of your proposed systems.
5. Submit answers in a pdf file format.

Type of SDLC - Waterfall Model

Prioritizing the core algorithmic structures is the best approach since our program first must be clearly defined and planned before any coding begins. A sequential phase-by-phase approach is needed to ensure our logic of the system's feature logic is fully interconnected to one another.

Task Responsibilities	
Verano, James Daniel	Full-Stack Developer
Braganza, Ralph Angelov	Backend Developer - Algorithms & Data Structure
Bal, Kevin Ranz	Backend Developer - Algorithms
Llaneta, Jhunel	Quality Assurance & Lead Tester
Uchi, Mao Sophie	System Architect

Proposed System 1 - Computer Lab Problem Log System (CLPSys)

1. How do you decide which computer problem should be repaired first?
2. How do you find a previous incident when you need to check its details?
3. How should the system display the history of incidents recorded for a specific laboratory computer? (Management)
4. How do you identify which computers experience the most problems?
5. What improvements would you want if the system were used regularly in the laboratory?
6. How can you verify that the system correctly finds, records, and updates incidents?
7. What happens if a technician accidentally updates the wrong incident or enters incorrect information?
8. When reviewing computer incidents, which information would you want to see first, and how would you prefer the records to be arranged?
9. Would grouping incidents into categories such as hardware, software, and network problems make it easier to manage them?
10. If the laboratory has hundreds of incident records, what difficulties would you expect when searching for a particular incident?

Proposed System 2 - Park Allocation Matrix System (PAMS)

1. What are the maximum physical dimensions of the garage (total floors, rows, and spaces per row)?
2. Can vehicles enter from multiple gate terminals all at once, or is there a single localized entrance?
3. Are certain spots strictly reserved for specific subscription vehicles (VIPs, handicapped, electric chargers)?
4. How does the management handle vehicles that lose their tickets or forget where they parked?
5. When pulling revenue logs, what is the default sorting format requested by accountants (by duration or fee)?
6. What is the frequency of vehicle entries during peak holiday or rush hours?
7. How are multi-car fleet accounts handled, and can multiple active vehicles share a single corporate code?
8. How to handle lanes due to maintenance work, or water leaks while searching for a parking slot?
9. In an emergency lockdown or fire event, what are the primary designated exit safety pathways?
10. Does the garage support real-time reservation bookings for cars that haven't arrived at the physical facility yet?

Proposed System 3 - PC Cafe System

1. What algorithm would you use for the queueing process?
2. How long will the customer have to wait for a vacant PC?
3. How will your system operate on the remaining balance of the customer?
4. How will your system broadcast the PC with the lowest remaining time left?
5. How will the customer know if there are any available PC's in the cafe?
6. What is the best approach when a customer wants to use a computer but there are malfunctioning PCs? How will you let them know?
7. How will the cashier find a customer's account if they forget their membership card and only remember their phone number?

8. How will your system know when the computer needs a PC maintenance?
9. If the customer wants to change computers, how will your system manage it?
10. How will the waiting line adjust itself if a customer pays extra for an express pass to skip ahead of everyone else?

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Google AI Overview**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Explanation of concepts.**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **To find out how to create a question that will answer a business process.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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